

Chapter 5

Operational Amplifiers

Op Amps

L#1	5.2	What is an Op Amp?
	5.3	Ideal Op Amp
L#2	5.4	Inverting
	5.5	non Inverting
L#3	5.6	Summing
	5.7	Difference Op Amp
L#4	5.8	Cascade circuit



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5.2 What is an Op Amp

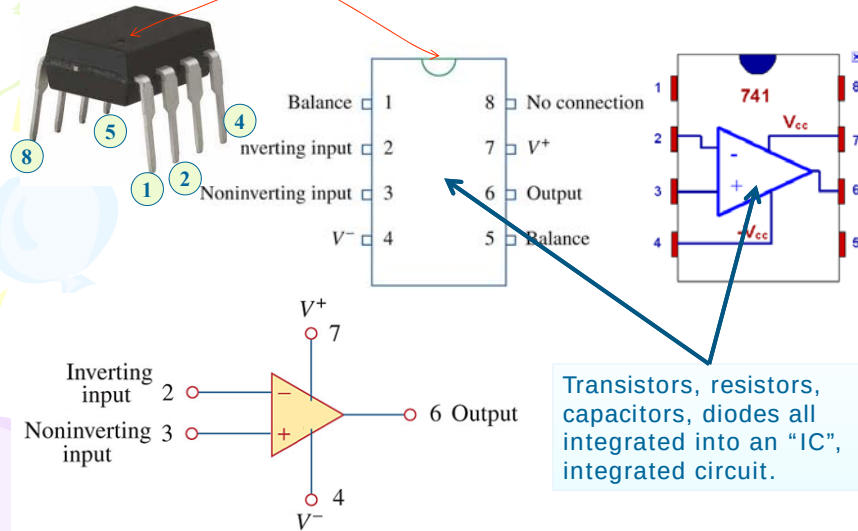
- It is an electronic unit that behaves like a **voltage-controlled voltage source**.
- It is an **active circuit element** designed to perform mathematical **operations** of ***addition, subtraction, multiplication, division, differentiation*** and ***integration***.
- Hence: **OPERATIONAL** Amplifier



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Pin Configuration and Circuit Symbol

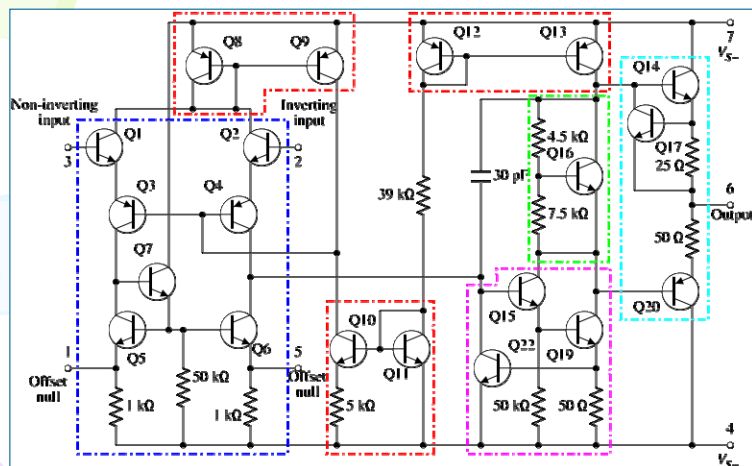


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Op Amp Circuit

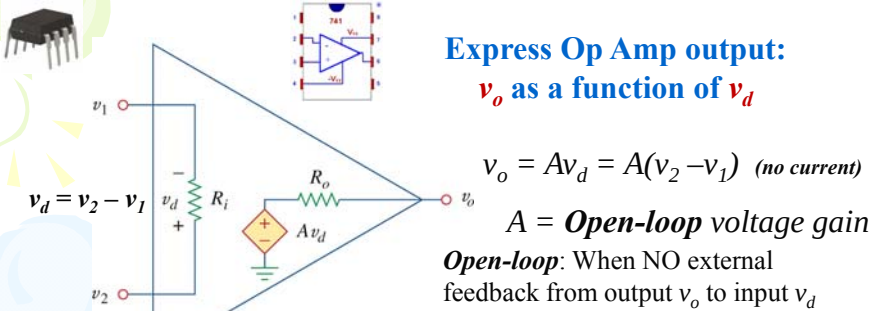


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Equivalent Circuit model of a non-ideal Op Amp



Typical ranges for op amp parameters

Parameter	Range	Ideal
A	$10^5 - 10^8$	∞
R_i	$10^5 - 10^{13} \Omega$	∞
R_o	$10 - 100 \Omega$	0
V_{CC}	$5 - 24 \text{ V}$	

Vcc = Supply voltage



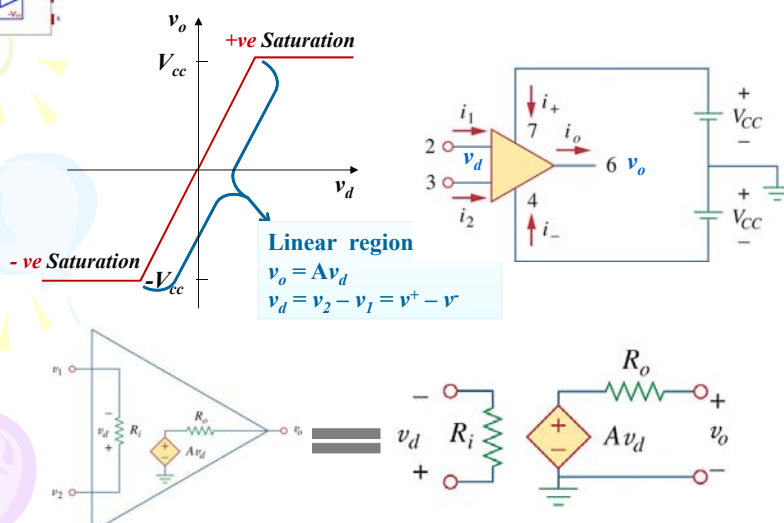
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Op-Amps Practical Limitation

Output cannot exceed $|V_{CC}|$

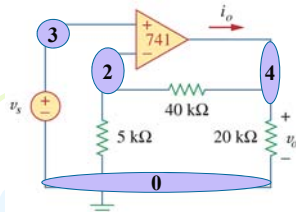


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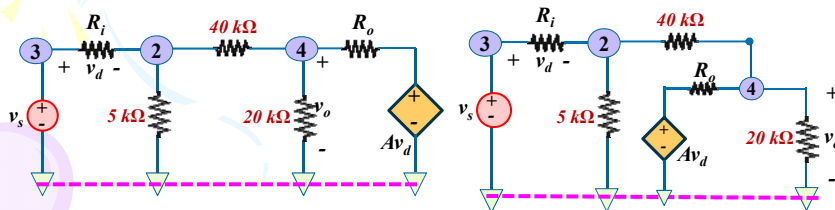
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PP 5.1 Use a 741 Op Amp with $A = 2 \times 10^5$, $R_i = 2 \text{ M}\Omega$, $R_o = 50 \Omega$. Determine the closed-loop gain v_o/v_s .



- Identify the Nodes
- Simplify the circuit



Actual/Practical Analysis



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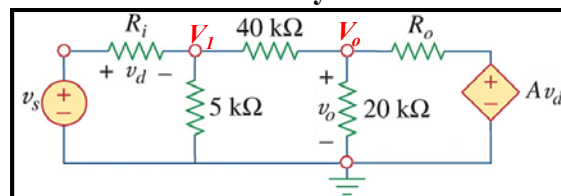
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PP 5.1 cont...

Use a 741 Op Amp with $A = 2 \times 10^5$, $R_i = 2 \text{ M}\Omega$, $R_o = 50 \Omega$. Determine the closed-loop gain v_o/v_s .

Nodal analysis:



KCL @ Node V_1

$$i_{R_i} + i_{5k} + i_{40k} = 0$$

$$\frac{V_1 - V_s}{2 \times 10^6} + \frac{V_1}{5 \times 10^3} + \frac{V_1 - V_o}{40 \times 10^3} = 0$$

$$V_1 - V_s + 400V_1 + 50V_1 - 50V_o = 0$$

$$451V_1 - 50V_o = V_s \quad (1)$$

KCL @ Node V_o

$$i_{40k} + i_{20k} + i_{R_o} = 0$$

$$\frac{V_o - V_1}{40 \times 10^3} + \frac{V_o}{20 \times 10^3} + \frac{V_o - Av_d}{50} = 0$$

$$V_o - V_1 + 2V_o + 800[V_o - A(V_s - V_1)] = 0$$

$$(-1 + 160 \times 10^6)V_1 + 803V_o - 160 \times 10^6V_s = 0$$

$$160 \times 10^6V_1 + 803V_o = 160 \times 10^6V_s \quad (2)$$

Calculate v_o when $v_s = 1 \text{ V}$

$$v_o \approx 9v_s$$

$$v_o \approx 9 \times 1 \approx 9 \text{ V}$$

Solve for $\frac{v_o}{v_s} =$

$$\frac{v_o}{v_s} \approx 9$$

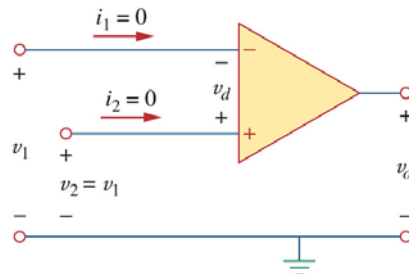


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5.3 Ideal Op-Amp



Since $R_i = \infty$ (open circuit)

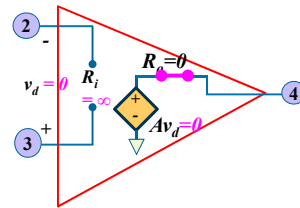
$$\therefore i_1 = i_2 = 0$$

$$v_d = v_2 - v_1 = 0$$

$$\therefore v_2 = v_1$$

An ideal op amp has the following characteristics:

1. Infinite open-loop gain, $A \approx \infty$
2. Infinite input resistance, $R_i \approx \infty$
3. Zero output resistance, $R_o \approx 0$



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PP 5.1 for Ideal op-amp

Assume $v_s = 1$ V and determine v_o and i_o .

v_o using actual analysis ≈ 9 V

Ideal Analysis

For Ideal Op-amp

$$i_1 = i_2 = 0$$

$$v^+ = v^-$$

$$v_s = v^+ = 1$$

$$v^- = v^+ = 1$$

$$\therefore v_d = v^+ - v^- = 0$$

KCL @ Node (2) or v^- to get

$$\frac{1}{5 \times 10^3} + \frac{1 - V_o}{40 \times 10^3} = 0$$

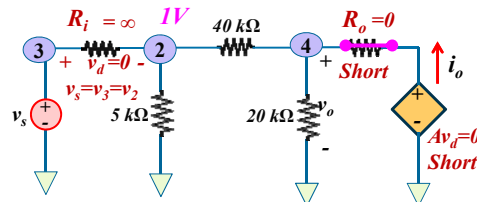
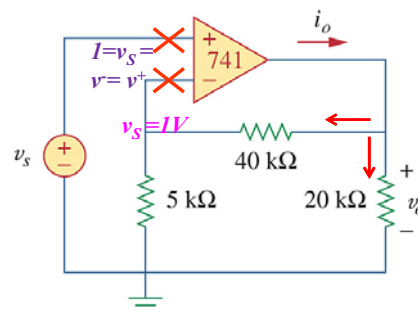
$$8 + 1 - V_o = 0$$

$$V_o = 9 \text{ V}$$

KCL @ Node (4) or v_o to get

$$i_o = \frac{9 - 1}{40 \times 10^3} + \frac{9}{20 \times 10^3}$$

$$i_o = 650 \mu\text{A}$$



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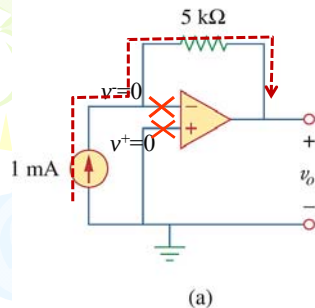
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Problem 5.8Determine v_o for both the circuits. (Ideal Op amp)*For Ideal Op-amp*

$i_1 = i_2 = 0$

$v^+ = v^-$



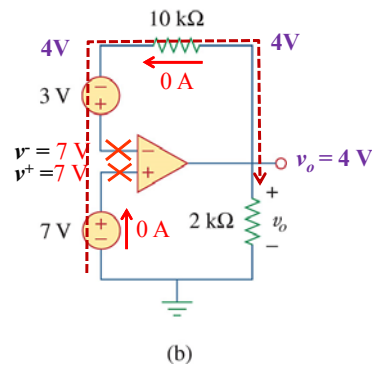
(a)

KCL @ v^- :

$I_{in} = I_{out}$

$$1 \times 10^{-3} = \frac{0 - v_o}{5 \times 10^3}$$

$$v_o = -5 \text{ V}$$



(b)

Or use KVL

$$-7 + 3 + V_{10k} + v_o = 0$$

$$-4 + 0 + v_o = 0$$

$$v_o = 4 \text{ V}$$

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Problem 5.9Determine v_o for both circuits*For Ideal Op-amp*

$i_1 = i_2 = 0$

$v^+ = v^-$

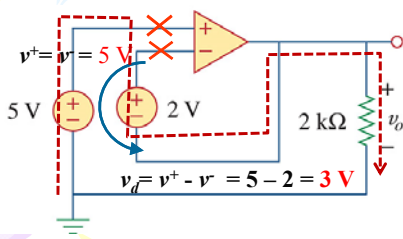
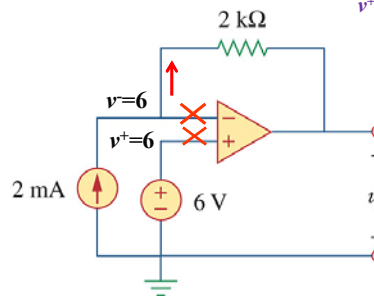
KCL @ v^- :

$I_{in} = I_{out}$

$$2 \times 10^{-3} = \frac{6 - v_o}{2 \times 10^3}$$

$$4 = 6 - v_o$$

$$v_o = 2 \text{ V}$$

Or use KVL

$$-5 + 2 + v_o = 0$$

$$v_o = 3 \text{ V}$$

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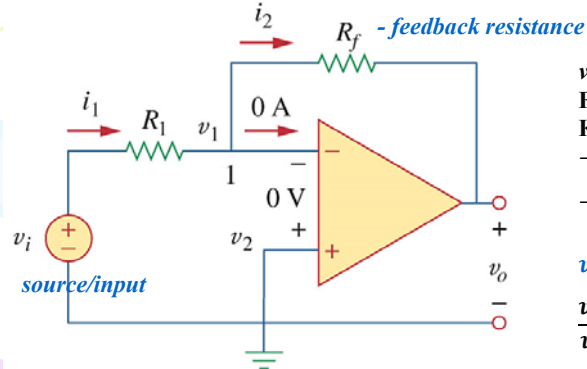
L#2 5.4 Inverting Amplifier

For Ideal Op-amp

$$i_1 = i_2 = 0$$

$$v^+ = v^-$$

- Inverting amplifier **reverses the polarity** of the input signal while amplifying it



$$v^+ = v^- = 0V \text{ or } v_1 = v_2$$

From first principle

KCL at v_1 or v^-

$$-i_1 + i_2 = 0$$

$$-\frac{v_i - 0}{R_1} + \frac{0 - v_o}{R_f} = 0$$

$$v_o = -\frac{R_f}{R_1} v_i$$

$$\frac{v_o}{v_i} = -\frac{R_f}{R_1}$$

$$-\frac{R_f}{R_1} = \text{Closed-loop gain}$$

Or Op-amp **GAIN**

Ideal Inverter:

- v^+ must be 0 V. ($v^+ = 0$ V)
- Only **ONE** source connected to v^-

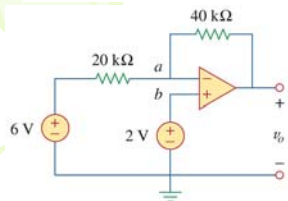


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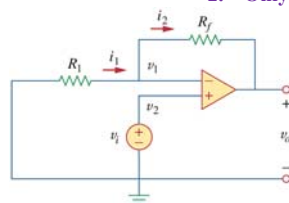
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Identify the Inverter



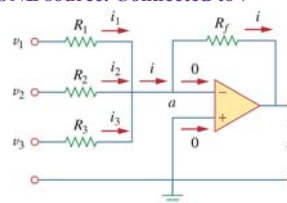
NOT Ideal Inverter. Why?

- NOT True. $v^+ = 2V$
- NOT True. More than one source.



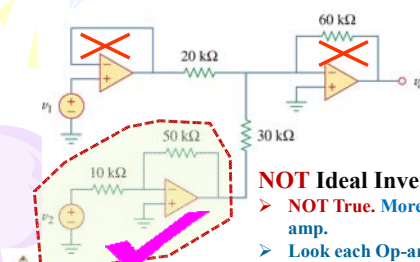
NOT Ideal Inverter. Why?

- NOT True. $v^+ = v_i$ and $v_i \neq 0$
- NOT True. NO source at v^- .



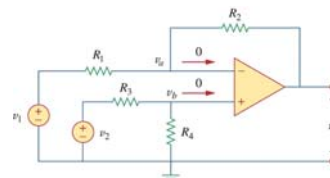
NOT Ideal Inverter. Why?

- True
- Three source at v^- . v_1 , v_2 & v_3



NOT Ideal Inverter. Why?

- NOT True. More than one Op-amp.
- Look each Op-amp individually



NOT Ideal Inverter. Why?

- NOT True. $v_b \neq 0$
- True



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PP 5.3 Determine the output voltage and current through the feedback resistor.

Is Ideal Inverter?

Yes

$$v_o = -\frac{R_f}{R_1} v_i$$

$$v_o = -\frac{120}{3} 30 \times 10^{-3}$$

$$v_o = -1.2 \text{ V}$$

From First principle:

KCL @ v^-

$$\frac{0 - 30\text{m}}{3\text{k}} + \frac{0 - v_o}{120\text{k}} = 0$$

$$-1200\text{m} - v_o = 0$$

$$v_o = -1.2 \text{ V}$$

Ohm's law

$$i_f = \frac{0 - v_o}{R_f}$$

$$i_f = \frac{0 - (-1.2)}{120 \times 10^{-3}} = 10 \mu\text{A}$$

Whenever NOT sure use **First principle**



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Example 5.4 Determine v_o .

Ideal Inverter

$$v_o = -\frac{R_f}{R_1} v_i$$

?

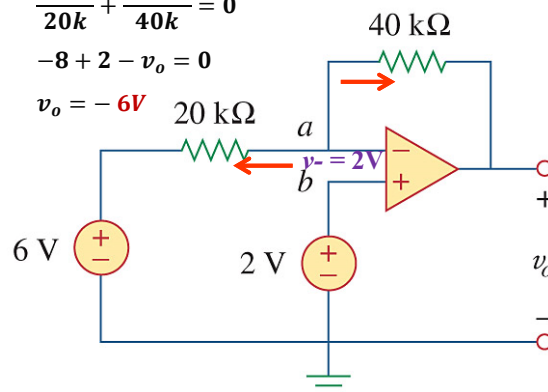
Can't apply classic equation because of the 2 V source. Solve from first principle

KCL @ a

$$\frac{2 - 6}{20\text{k}} + \frac{2 - v_o}{40\text{k}} = 0$$

$$-8 + 2 - v_o = 0$$

$$v_o = -6 \text{ V}$$



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5.4 Inverting Amplifier

Special Case: Current-to-Voltage converter

For Ideal Op-amp
 $i_1 = i_2 = 0$
 $v^+ = v^-$

KCL at v^- : $i_s = \frac{0 - v_o}{R}$

$v_o = -i_s R$

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Problem 5.19

Calculate i_o .

Can the shaded circuit be replaced by a source and resistor?

?

Yes

Source Transformation

Is the circuit ideal Inverter?

Yes

Not sure!!

For Ideal Op-amp
 $i_1 = i_2 = 0$ ✓
 $v^+ = v^- = 0$ ✓

KCL @ i_o

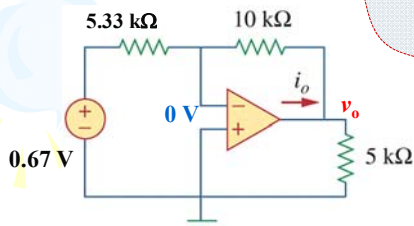
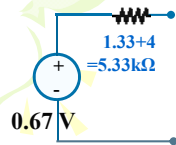
$$i_o = i_f + i_{5k}$$

$$= \frac{v_o - 0}{10k} + \frac{v_o}{5k}$$

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Problem 5.19Calculate i_o .*Ideal Inverter*

$$v_o = -\frac{10}{5.33} 0.67$$

$$= -1.25 \text{ V}$$

Is the circuit ideal Inverter?

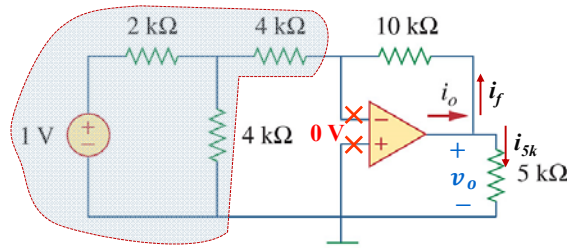
Yes

Not sure!!

For Ideal Op-amp

$$i_1 = i_2 = 0$$

$$v^+ = v^- = 0$$



$$i_o = i_f + i_{5k}$$

$$= \frac{v_o - 0}{10k} + \frac{v_o}{5k}$$

$$= 3 \frac{v_o}{10k}$$

$$i_o = 3 \frac{-1.25}{10k}$$

$$= -375 \mu\text{A}$$

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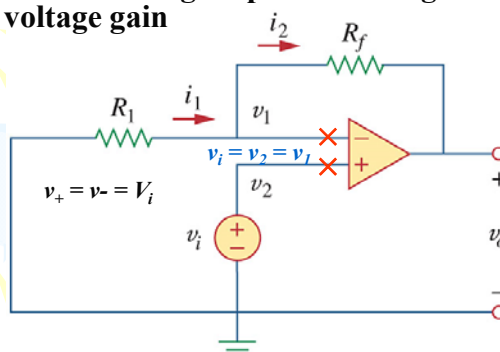
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5.5 Non-inverting Amplifier*For Ideal Op-amp*

$$i_1 = i_2 = 0$$

$$v_+ = v_-$$

- Non-inverting amplifier is designed to produce **positive** voltage gain

From first principle
KCL at v_i :

$$i_1 = i_2$$

$$\frac{0 - v_i}{R_1} = \frac{v_i - v_o}{R_f}$$

$$-R_f v_i = R_1 v_i - R_1 v_o$$

$$v_o = \left(\frac{R_1 + R_f}{R_1} \right) v_i$$

$$v_o = \left(1 + \frac{R_f}{R_1} \right) v_i$$

$$\frac{v_o}{v_i} = \left(1 + \frac{R_f}{R_1} \right)$$

called closed-loop gain

Ideal non-Inverter:

- Only source connected to v^+
- $R_1 \neq 0$

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Identify the Op-amp

NOT Inverter or non-inverter. Why?
1. More than one source.

Ideal Inverter:
1. v^+ must be 0 V. ($v^+ = 0$ V)
2. Only ONE source. Connected to v^-

Non-Inverter. Why?
1. True. Only source connected to v^+
2. True. $R_f \neq 0$

Ideal non-Inverter:
1. Only source connected to v^+
2. $R_f \neq 0$

NOT Inverter or non-inverter. Why?
1. More than one source.

Non-inverter

inverter

NOT Inverter or non-inverter. Why?
1. More than one Op-amp.
2. Look each Op-amp individually

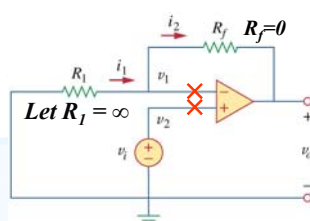
NOT Inverter or non-inverter. Why?
1. More than one source.

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5.5 Non-inverting Amplifier

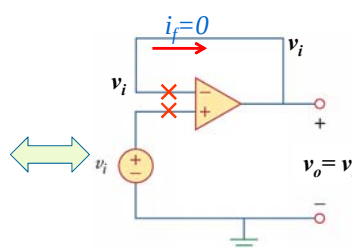
Special Case: Voltage Follower



$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

$$v_o = v_i$$

$$V_{\text{Output}} = V_{\text{Input}}$$



For Ideal Op-amp

$$i_1 = i_2 = 0$$

$$v_+ = v_-$$

- Advantage: v_i sources a very low current while v_o can source high current. (Remember Op Amp is an Active element getting energy via V_{CC} and $-V_{CC}$).
- Circuit is used as a BUFFER to isolate one stage of a circuit from another.

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PP5.5 Calculate v_o . Is the circuit Inverter/non-inverter?
Non-inverter, bc only source at v^+
Not sure!!

For Ideal Op-amp
 $i_1 = i_2 = 0$
 $v^+ = v^-$

$i^+ = 0 A$
 Because of this, the 3V is divided across the 4 kΩ and 8 kΩ resistors, hence a basic voltage divider.

Voltage divider

$$v^+ = 3 \frac{8}{4 + 8} = 2 V = v^-$$

KCL @ v^-

$$i_{2k} + i_{5k} = 0$$

$$\frac{2}{2k} + \frac{2 - v_o}{5k} = 0$$

$$10 + 4 - 2v_o = 0$$

$$v_o = 7 V$$

Or
$$v_o = \left(1 + \frac{R_f}{R_1}\right) v^+$$

$$= \left(1 + \frac{5k}{2k}\right) 2$$

$$v_o = 7 V$$

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Problem 5.25 Calculate v_o .

For Ideal Op-amp
 $i_1 = i_2 = 0$
 $v^+ = v^-$

Or
Voltage Follower
 Voltage divider

$$v_o = 2 \frac{20k}{12k + 20k}$$

$$= 1.25 V$$

Is the circuit Inverter/non-inverter?
Non-inverter, bc only source at v^+

$$v_{o1} = \left(1 + \frac{R_f}{R_1}\right) v_i$$

$$= \left(1 + \frac{0}{\infty}\right) 2$$

Voltage divider

$$v_o = v_{o1} \frac{20}{2 + 20}$$

$$= 1.25 V$$

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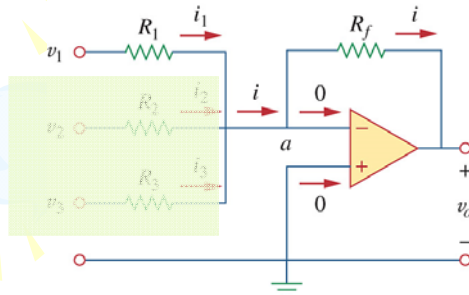
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Ch. 5 L#3

5.6 Summing Amplifier

Is the circuit Inverter/non-inverter?

Neither. Has Three source: v_1, v_2, v_3



For Ideal Op-amp

$$i_1 = i_2 = 0$$

$$v^+ = v^-$$

Using Superposition

$$v_o = v_{o1} + v_{o2} + v_{o3}$$

$$v_{o1} = -\frac{R_f}{R_1} v_1 \quad \text{- inverter}$$

$$v_{o2} = -\frac{R_f}{R_2} v_2$$

$$v_{o3} = -\frac{R_f}{R_3} v_3$$

$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3\right)$$

- Summing Amplifier is an op amp circuit that combines several inputs and produces an output that is the **weighted sum** of the inputs.
- Basic configuration of an inverting amplifier but with **multiple inputs** at v^-



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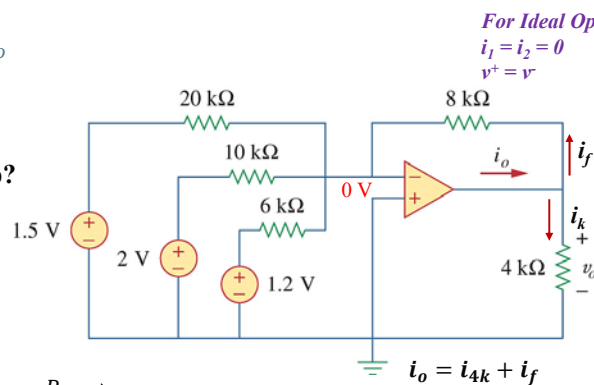
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PP 5-6

Determine v_o and i_o for the following circuit:

Identify the op-amp?

Summing op-amp



For Ideal Op-amp

$$i_1 = i_2 = 0$$

$$v^+ = v^-$$

$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3\right)$$

$$v_o = -\left(\frac{8}{20} 1.5 + \frac{8}{10} 2 + \frac{8}{6} 1.2\right)$$

$$v_o = -(0.6 + 1.6 + 1.6)$$

$$= -3.8 \text{ V}$$

$$i_o = i_{4k} + i_f$$

$$= \frac{v_o}{4k} + \frac{v_o - 0}{8k}$$

$$= \frac{-3.8}{4k} + \frac{-3.8}{8k}$$

$$= -1.425 \text{ mA}$$



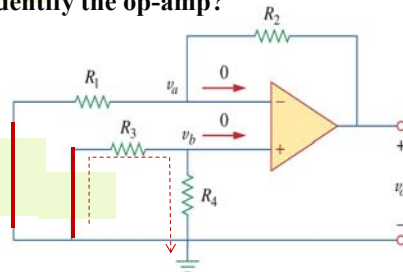
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5.7 Difference Amplifier

Identify the op-amp?



Using Superposition

$$v_o = v_{o,v1} + v_{o,v2}$$

Zero v_2

Although $R3 \parallel R4$, but No current is flowing

$v_b = 0$; Circuit becomes **Inverter**

$$v_{o,v1} = -\left(\frac{R_2}{R_1}\right) v_1$$

Not: Inverter;

non-inverter;
summing amplifier.

Has input/source at both v^+ & v^-

Zero v_1

Only v^+ has a source;

non-inverter

$$v_{o,v2} = \left(1 + \frac{R_2}{R_1}\right) v_b \quad \text{Note: its NOT } v_2$$

v_b = voltage across R_4

$$v_b = v^+ = v_a$$

$$v_b = \left(\frac{R_4}{R_3 + R_4}\right) v_2 \quad \text{voltage divider}$$

$$v_b = \left(\frac{1}{1 + \frac{R_3}{R_4}}\right) v_2$$

$$v_{o,v2} = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{1}{1 + \frac{R_3}{R_4}}\right) v_2$$

$$v_o = \frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} v_2 - \frac{R_2}{R_1} v_1$$

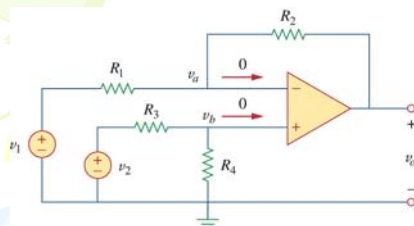


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5.7 Difference Amplifier



$$v_b = v^+ = v_a$$

$$v_o = \frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} v_2 - \frac{R_2}{R_1} v_1$$

1. If $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ then

$$v_o = \frac{R_2}{R_1} (v_2 - v_1);$$

$$\text{if } v_1 = v_2 \Rightarrow v_o = 0$$

2. If $R_2 = R_1$ and $R_3 = R_4$ or $\frac{R_1}{R_2} = \frac{R_3}{R_4} = 1$

$$\Rightarrow v_o = v_2 - v_1$$

- Difference amplifier is a device that amplifies the **difference between two inputs** but rejects any signals common to the two inputs.
- **Combination of the inverting and non-inverting op amps.**



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Example 5.7

Design an Op-Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$

$$v_o = \frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} v_2 - \frac{R_2}{R_1} v_1$$

Design 1: Using only 1 Op-Amp

Since there are two source v_1 & v_2 start with Difference op-amp

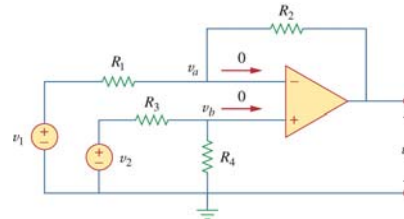
$$-\frac{R_2}{R_1} v_1 = -5v_1$$

$$\therefore R_2 = 5R_1$$

Theoretically any possible value can be chosen.

Let: $R_1 = 1 \text{ k}\Omega$; $\Rightarrow R_2 = 5 \text{ k}\Omega$

For practical design opt for available resistors with low power ratings.



$$\frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} v_2 = 3v_2 \quad \frac{5 \left(1 + \frac{1}{5}\right)}{1 \left(1 + \frac{R_3}{R_4}\right)} = 3$$

$$\Rightarrow 5 \left(1 + \frac{1}{5}\right) = 3 \left(1 + \frac{R_3}{R_4}\right)$$

$$1 = R_3/R_4 \Rightarrow R_3 = R_4$$

Choose: 5 kΩ

Theoretically any value can be chosen



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Example 5.7

Design an Op-Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$

Design 2: Using only Summing Op-Amp

Since there are two source v_1 & v_2

$$v_o = -\frac{R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2$$

Calculate R_1 , R_2 & R_f by Comparing the two equations

$$-\frac{R_f}{R_1} v_1 = -5v_1 \Rightarrow R_f = 5R_1$$

$$-\frac{R_f}{R_2} v_2 = 3v_2 \Rightarrow R_f = -3R_2$$

Since R can not be -ve, $v_2 = -ve$

$$\Rightarrow 5R_1 = 3R_2$$

$$\Rightarrow R_2 = \frac{5}{3} R_1$$

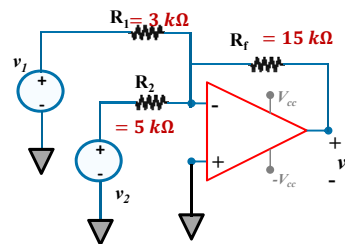
Theoretically any possible value can be chosen.

Let: $R_1 = 3 \text{ k}\Omega$;

$$\Rightarrow R_2 = 5 \text{ k}\Omega$$

$$R_f = 5R_1 = 5 \times 3 \text{ k} = 15 \text{ k}\Omega$$

For practical design opt for available resistors with low power ratings.



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Example 5.7

Design an Op Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$

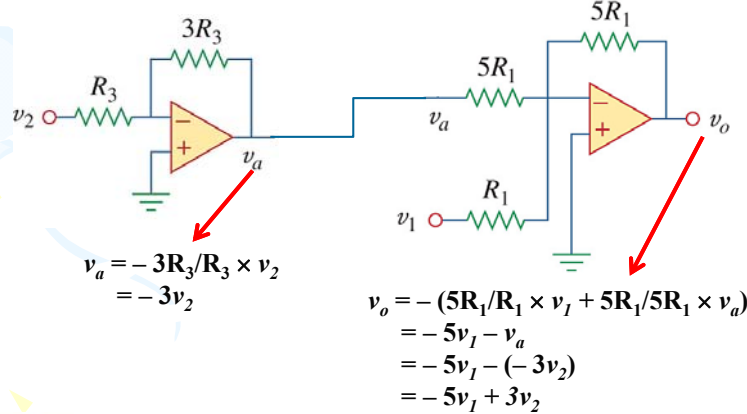
For Ideal Op-amp

$$i_1 = i_2 = 0$$

$$v_+ = v_-$$

Design 3: Using 2 Op Amps

1. Build a ~~non~~-inverting amp to obtain $v_a = -3v_2$.
2. Feed the output v_a into a summing amplifier along with v_1 .



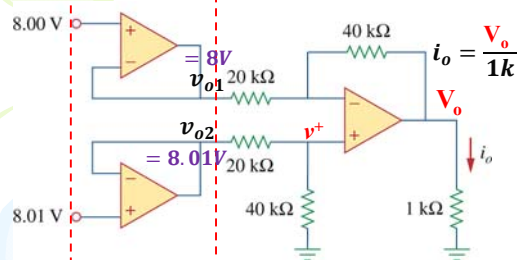
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PP 5-8

Determine i_o for the following circuit:



Voltage Followers

Difference Amplifier

From first principles:

$$v_+ = v_- = 8.01 \frac{40}{60} = 5.34V$$

$$\text{KCL @ } v^-: \frac{5.34 - 8}{20k} + \frac{5.34 - V_o}{40k} = 0$$

$$V_o = +0.02 V$$

$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

$$\therefore i_o = \frac{0.02}{1k} = 20 \text{ mA}$$

Using Classical equation

$$v_o = \frac{40(1 + 20/40)}{20(1 + 20/40)} v_2 - \frac{40}{20} v_1$$

$$= 2v_2 - 2v_1$$

$$= 2(8.01 - 8)$$

$$= 0.02 V$$

$$i_o = \frac{0.02}{1k} = 20 \text{ mA}$$



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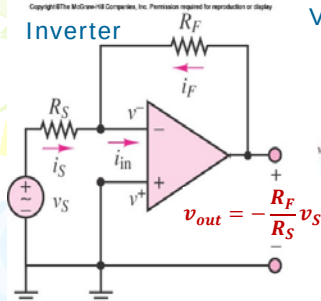
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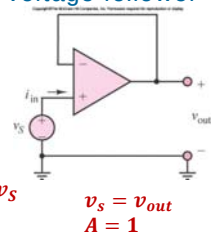
Summary: Ideal Op-Amp

$$i_{in} = 0 \text{ and } v^- = v^+$$

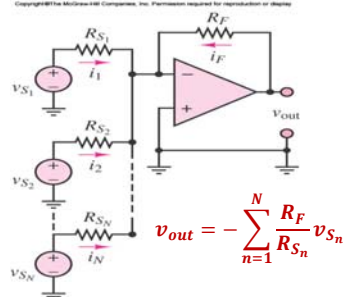
Inverter



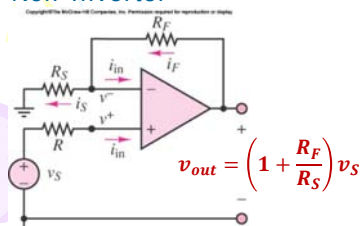
Voltage follower



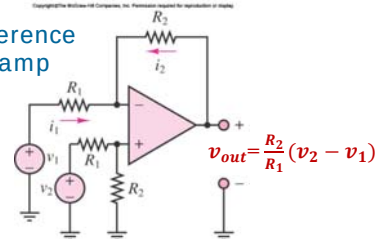
Summer Op-amp



Non-Inverter

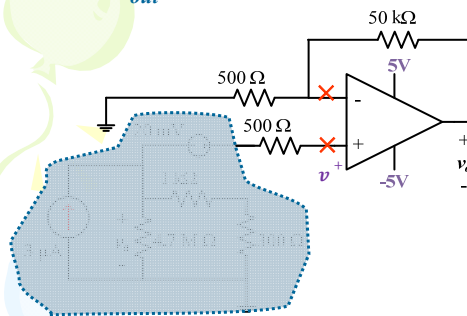


Difference Op-amp



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Find v_{out} ?



Is the circuit Inverter/non-inverter?

Not sure!!

Non-inverter, bc only source at v^+

$$v_{out} = \left(1 + \frac{R_f}{R_1}\right) v^+$$

$$v^+ = v_a - 20m$$

KCL @ v_a

$$3\mu = \frac{v_a}{4.7M} + \frac{v_a}{1000 + 300}$$

NO current flows through 500Ω

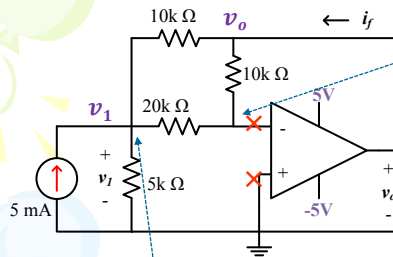
$$v_a = 3.9mV$$

$$= 3.9m - 20m = -16.1mV$$

$$v_{out} = \left(1 + \frac{50k}{500}\right) (-16.1m)$$

$$v_{out} = -1.63V$$

Calculate v_o and i_f .



From first principle:

KCL @ v_1 :

$$5\text{mA} = i_{5k} + i_{20k} + i_{10k}$$

$$5\text{mA} = \frac{v_1}{5k} + \frac{v_1 - v_o}{20k} + \frac{v_1 - v_o}{10k}$$

$$v^+ = 0 = v^-$$

$$7v_1 - 2v_o = 100 \quad (1)$$

KCL @ v^- :

$$i_{20k} + i_{10k} = 0$$

$$-v_1 - 2v_o = 0 \quad (2)$$

Solve

$$v_o = -6.25\text{V}$$

$$v_1 = 12.5\text{V}$$

$$i_f = i_{10k} + i_{10k}$$

$$= \frac{v_o - v_1}{10k} + \frac{v_o - v^-}{10k}$$

$$i_f = -2.5\text{mA}$$



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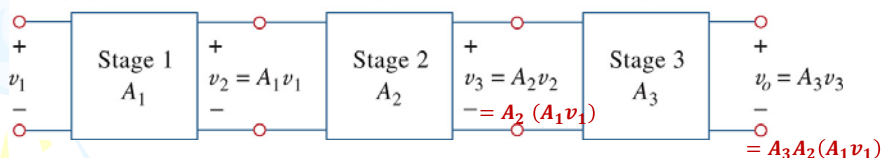
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Ch. 5 L# 4

5.8 Cascaded Op Amp Circuits

- Consider we want an Op-amp that can have gain of 1000?
- Cascading**
- It is a head-to-tail arrangement of two or more op amp circuits such that the output of one is the input to the next.



$$v_o = A_3 v_3 = A_3 (A_2 v_2) = A_3 A_2 (A_1 v_1)$$

$$\text{Total Amplification } A = A_1 A_2 A_3$$

$$\text{Consider } A_1 = A_2 = A_3$$

$$A = 1000$$

$$A_1 = \sqrt[3]{1000} = 10$$



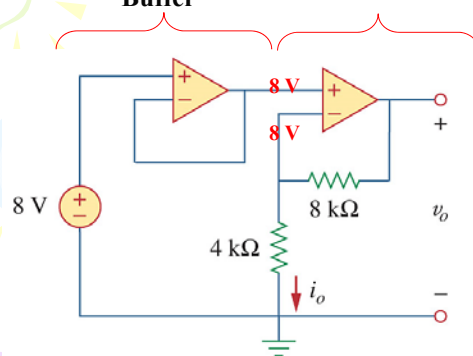
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PP 5-9Determine v_o and i_o .Voltage
Follower or
Buffer

Non-Inverter



$$v_+ = v_- = 8 \text{ V}$$

KCL at v_-

$$\frac{8 - 0}{4k} + \frac{8 - v_o}{8k} = 0$$

$$16 + 8 = v_o$$

$$v_o = 24 \text{ V}$$

$$i_o = \frac{v_-}{4k} = \frac{8}{4k}$$

$$= 2 \text{ mA}$$

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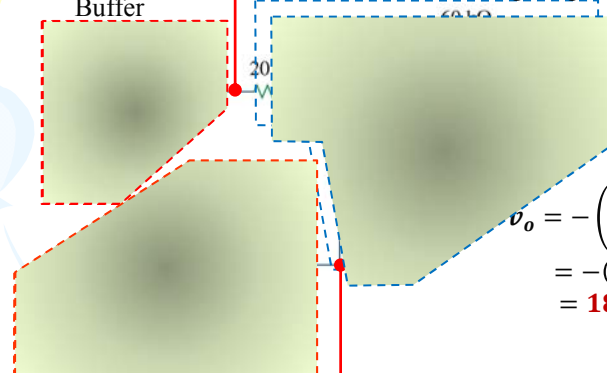
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PP 5-10If $v_1 = 4 \text{ V}$ and $v_2 = 3 \text{ V}$, determine v_o .

$$v_a = v_1 = 4 \text{ V}$$

Buffer

Summing Amp



$$v_o = -\left(\frac{60}{20}v_a + \frac{60}{30}v_b\right)$$

$$= -(3 \times 4 + 2 \times -15)$$

$$= 18 \text{ V}$$

$$v_b = -\frac{50}{10}v_2 = -15 \text{ V}$$

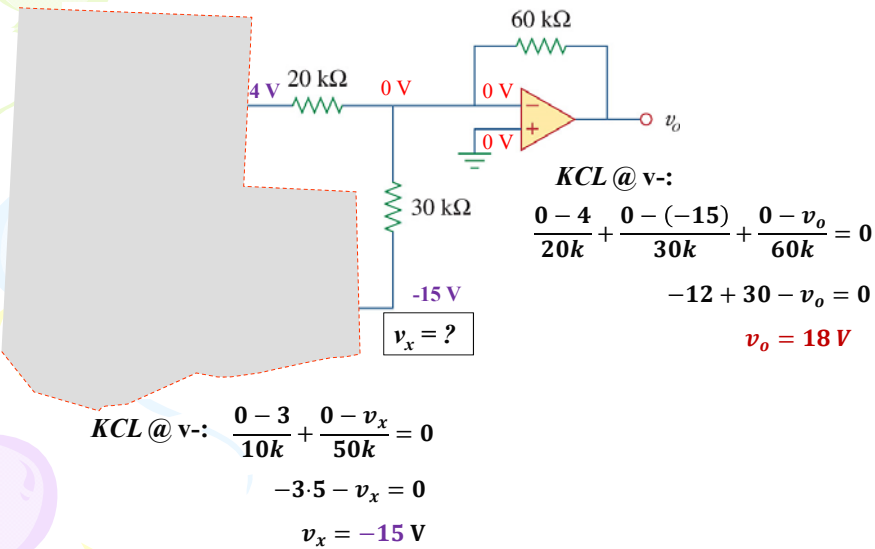
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PP 5-10

If $v_1 = 4\text{ V}$ and $v_2 = 3\text{ V}$, determine v_o . From first principles:



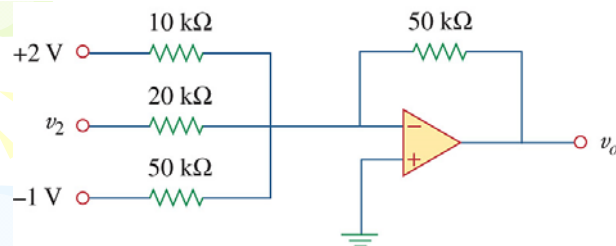
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Problem 5-39

Determine v_2 such that $v_o = -16.5\text{ V}$.



$$v_o = -\left(\frac{50}{10}2 + \frac{50}{20}v_2 + \frac{50}{50}(-1)\right) = -16.5$$

$$-10 + 1 - 2.5v_2 = -16.5$$

$$-2.5v_2 = -7.5$$

$$v_2 = 3\text{ V}$$



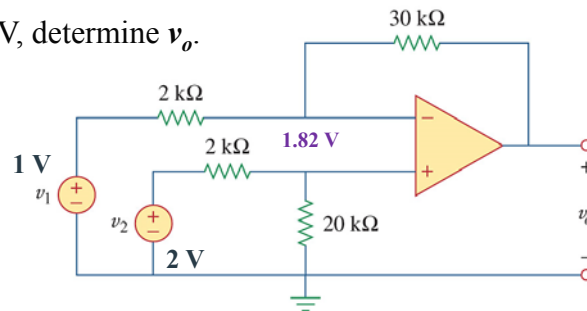
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Problem 5-39

If $v_1 = 1\text{ V}$ and $v_2 = 2\text{ V}$, determine v_o .

**First Principles**

$$v_+ = \frac{20}{20+2} v_2 = 1.81\text{ V} = v_-$$

KCL @ v_- :

$$\frac{1.81 - 1}{2k} + \frac{1.81 - v_o}{30k} = 0$$

$$12.27 + 1.81 - v_o = 0$$

$$v_o = 14.09\text{ V}$$

$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$



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Problem 5-59

Determine v_o/v_s if $R = 20\text{ k}\Omega$.

Cascading

Or

$$v_a = \left(1 + \frac{3R}{R}\right) v_s = 4v_s$$

$$v_o = -\frac{4R}{R} v_a = -4v_a$$

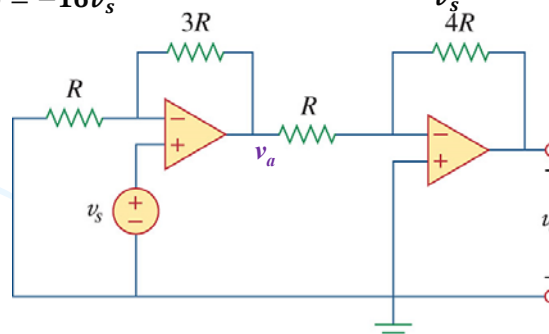
$$= -4(4v_s) = -16v_s$$

$$\frac{v_o}{v_s} = -16$$

$$A_1 = 1 + \frac{3R}{R} = 4$$

$$A_2 = -\frac{4R}{R} = -4$$

$$A = A_1 A_2 = \frac{v_o}{v_s} = -16$$

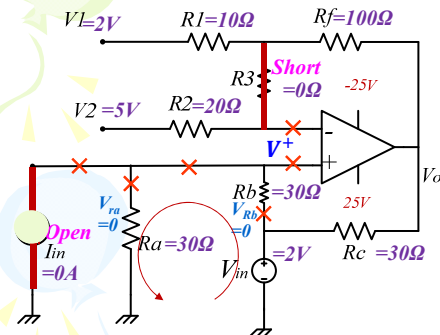


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Ch. 5 L#5 Revision



$$V^+ = 2V$$

$$\text{KCL@V}^-: i_{20} + i_{10} + i_{100} = 0$$

$$\frac{2-5}{20} + \frac{2-2}{10} + \frac{2-V_o}{100} = 0$$

$$V_o = -13V$$

Calculate the output voltage V_o .

Given $R1=10\Omega$; $R2=20\Omega$; $R3=0\Omega$;

$Rf=100\Omega$; $V1=2V$; $V2=5V$;

$Ra=Rb=Rc=30\Omega$, $I_{in}=0A$; $V_{cc}=\pm 25V$

and $V_{in}=2V$.

$$V^+ = V_{Ra} = \frac{30}{30+30} * 2 = 1V$$

$$\text{KCL@V}^-: i_{20} + i_{10} + i_{100} = 0$$

$$\frac{1-5}{20} + \frac{1-2}{10} + \frac{1-V_o}{100} = 0$$

$$V_o = -29V$$

Is this the answer?

Sure!

Wrong

Op Amp saturated

$V_{cc}=\pm 25V$

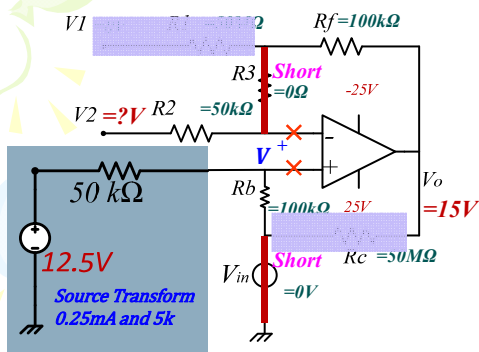
$$\therefore V_o = -25V$$



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Given $R1=50M\Omega$; $R2=50k\Omega$; $R3=0\Omega$; $Rf=100k\Omega$; $V1=0V$; $V_{in}=0V$;
 $I_{in}=0.25mA$; $Ra=50k\Omega$; $Rb=100k\Omega$ and $Rc=50M\Omega$. For what value of $V2$,
 $V_o=15V$.



$50M\Omega$ is very large value, for ideal op-amp it is safe to assume NO current flows through it. ($i_{50M} = 0$)

$$V^+ = V_{50k} = \frac{100}{50+100} 12.5$$

$$= 8.33V$$

$$\text{KCL@V}^-: i_{50k} + i_{50M} + i_{100k} = 0$$

$$\frac{8.33-V2}{50k} + \frac{8.33-0}{50M} + \frac{8.33-V_o}{100k} = 0$$

$$\frac{8.33-V2}{50k} + 167 \times 10^{-9} + \frac{8.33-V_o}{100k} = 0$$

$$\therefore V_2 = 5V$$

Difference amplifier

$$v_{out} = \frac{R_2}{R_1} (V_{in} - V_2)$$

$$= \frac{100}{50} (12.5 - V_2) = 15$$

$$\Rightarrow V_2 = 5V$$



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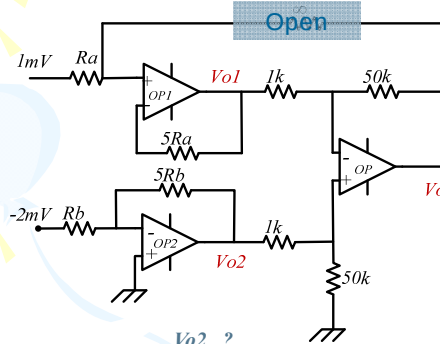
Calculate Vo1, Vo2 and Vo

$V_{o1} ?$

$R_f = \infty;$

OP1: Voltage Follower

$V_{o1} = 1 \text{ mV}$



$V_o ?$

OP: Ideal Difference amplifier

$$V_o = \frac{50k}{1k} (V_{o2} - V_{o1}) = 50 * 9m$$

$$= 450 \text{ mV}$$

$V_{o2} ?$

OP2: Inverter

$$V_{o2} = -\frac{5Rb}{Rb} * -2m$$

$$= 10 \text{ mV}$$



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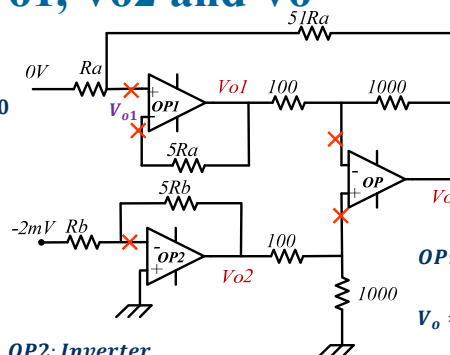
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Calculate Vo1, Vo2 and Vo

OP1: KCL @ V^+

$$\frac{V_{o1} - 0}{Ra} + \frac{V_{o1} - V_o}{51Ra} = 0$$

$$52V_{o1} = V_o \quad (1)$$



OP2: Inverter

$$V_{o2} = -\frac{5Rb}{Rb} * -2m$$

$$= 10m$$

OP: Difference amp

$$V_o = \frac{1000}{100} (V_{o2} - V_{o1})$$

$$= 10(10m - V_{o1}) \quad (2)$$

Sub (1) into (2)

$$52V_{o1} = 100m - 10V_{o1}$$

$$V_{o1} = \frac{100m}{62} = 1.63mV$$

$$V_o = 52 * 1.63m$$

$$= 83.87mV$$



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